

Attention-Enhanced U-Net Road Segmentation Project

Code Location:

<https://github.com/ajanakiraman2024/attention-enhanced-unet-segmentation>

Project Overview

This project implements an Attention-Enhanced U-Net architecture for road segmentation from satellite imagery. The model uses attention gates to focus on relevant features and improve segmentation accuracy compared to standard U-Net.

Project Structure

```
attention-enhanced-unet-segmentation/
|
+-- config.py                # Configuration settings
+-- train.py                 # Training script
+-- evaluate.py              # Evaluation script
+-- inference.py             # Inference/prediction script
+-- requirements.txt         # Python dependencies
+-- README.md                # Project documentation
|
+-- models/                  # Neural network architectures
|   +-- __init__.py
|   +-- unet.py              # Standard U-Net model
|   +-- attention_unet.py    # Attention U-Net model
|
+-- utils/                   # Utility functions
|   +-- __init__.py
|   +-- dataset.py           # Data loading & preprocessing
|   +-- losses.py            # Loss functions
|   +-- metrics.py           # Evaluation metrics
|
+-- notebooks/               # Jupyter notebooks
|   +-- UNet_Road_Segmentation_Colab.ipynb
|
+-- data/                    # Dataset
|   +-- training/
|       |   +-- images/      # Original images (100 PNG)
|       |   +-- groundtruth/ # Ground truth masks (100 PNG)
|       |   +-- images_generated/ # Augmented images (1003 PNG)
|       |   +-- groundtruth_generated/ # Augmented masks (1003 PNG)
|   +-- test_set_images/     # Test data (50 images)
|
+-- images/                  # Visualization outputs
+-- outputs/                 # Model checkpoints
```